Top-K Error-Feedback SGD (EF-SGD) with Sparse Gradients: Algorithm and Convergence Theory

Algorithm Name

Top-K Error-Feedback Stochastic Gradient Descent (Top-K EF-SGD)

Mathematical Formulation

We aim to minimize a possibly non-convex, L-smooth objective

$$\min_{w \in \mathbb{R}^d} f(w) = \mathbb{E}_{z \sim \mathcal{D}}[F(w; z)].$$

At iteration t, we draw a sample $z_t \sim \mathcal{D}$, compute a stochastic gradient $g_t = \nabla F(w_t; z_t)$, and apply a sparse compressor $\mathcal{C}$ that keeps only the K largest-magnitude coordinates (Top-K) together with error-feedback.

- Error-feedback state $e_t \in \mathbb{R}^d$ accumulates past compression errors. - Step-size (learning rate) $\eta_t > 0$.

Update equations:

$$\begin{aligned} u_t &= e_t + \eta_t g_t, \\ s_t &= \mathcal{C}(u_t) \quad \text{(Top-}K\text{)}, \\ w_{t+1} &= w_t - s_t, \\ e_{t+1} &= u_t - s_t. \end{aligned}$$

Equivalently, $s_t = \eta_t g_t + e_t - e_{t+1}$ and

$$w_{t+1} = w_t - \eta_t g_t - e_t + e_{t+1}.$$

Top-K compressor: - For $x \in \mathbb{R}^d$, $\mathrm{TopK}(x)$ keeps the K coordinates of x with largest absolute values and sets others to zero.

Contractive property of Top-K (deterministic):

$$\|x - \mathrm{TopK}(x)\|^2 \le (1 - \tau)\|x\|^2, \quad \tau := \frac{K}{d} \in (0, 1].$$

Pseudocode

Inputs: initial $w_1 \in \mathbb{R}^d$, horizon T, step sizes $\{\eta_t\}_{t=1}^T$, sparsity K.
Initialize: $e_1 \leftarrow 0 \in \mathbb{R}^d$.
For $t = 1, 2, \dots, T$:

1. Sample data point $z_t \sim \mathcal{D}$ and compute $g_t \leftarrow \nabla F(w_t; z_t)$.

2. Form accumulator $u_t \leftarrow e_t + \eta_t g_t$.

3. Compress $s_t \leftarrow \mathrm{TopK}(u_t)$ (keep K largest $|u_t|$ entries).

4. Update parameters $w_{t+1} \leftarrow w_t - s_t$.

5. Update error $e_{t+1} \leftarrow u_t - s_t$.

Output: (optionally) a random iterate w_R with R uniform in $\{1, \ldots, T\}$, or the average $\bar{w}_T = \frac{1}{T}\sum_{t=1}^{T} w_t$.

Dry Run Example

Objective: $f(w) = (w-3)^2$ (1D), gradient $\nabla f(w) = 2(w-3)$. Let $w_1 = 0$, $\eta_t \equiv \eta = 0.1$, $K = d = 1$ (no sparsification loss), $e_1 = 0$.

Iteration $t = 1$:

$$g_1 = 2(0-3) = -6, \quad u_1 = 0+0.1(-6) = -0.6, \quad s_1 = -0.6, \quad w_2 = 0-(-0.6) = 0.6, \quad e_2 = u_1 - s_1 = 0.$$

Objective: $f(w_2) = (0.6-3)^2 = 5.76$.

Iteration $t = 2$:

$$g_2 = 2(0.6-3) = -4.8, \quad u_2 = 0+0.1(-4.8) = -0.48, \quad s_2 = -0.48, \quad w_3 = 0.6-(-0.48) = 1.08, \quad e_3 = 0.$$

Objective: $f(w_3) = (1.08-3)^2 = 3.6864$.

Iteration $t = 3$:

$$g_3 = 2(1.08-3) = -3.84, \quad u_3 = -0.384, \quad s_3 = -0.384, \quad w_4 = 1.08-(-0.384) = 1.464, \quad e_4 = 0.$$

Objective: $f(w_4) = (1.464-3)^2 = 2.356896$.

Assumptions

Let $\{\mathcal{F}_t\}$ be the filtration generated by the algorithmic randomness up to time t (including w_t, e_t).

A1 Smoothness: f is L-smooth: for all x, y, $f(y) \le f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\|y-x\|^2$.

A2 Bounded below: $f^* := \inf_w f(w) > -\infty$ and $\Delta_f := f(w_1) - f^* < \infty$.

A3 Unbiased stochastic gradients: $\mathbb{E}[g_t \mid \mathcal{F}_t] = \nabla f(w_t)$.

A4 Variance bound: $\mathbb{E}[\|g_t - \nabla f(w_t)\|^2 \mid \mathcal{F}_t] \le \sigma^2$. Consequently,

$$\mathbb{E}[\|g_t\|^2 \mid \mathcal{F}_t] \le \|\nabla f(w_t)\|^2 + \sigma^2.$$

A5 Compressor contractivity (Top-K): for all $x \in \mathbb{R}^d$,

$$\|x - \mathcal{C}(x)\|^2 \le (1-\tau)\|x\|^2, \quad \tau = K/d \in (0, 1].$$

Main Convergence Theorem

Let Assumptions A1–A5 hold and let $\eta_t \equiv \eta$ satisfy

$$0 < \eta \le \min\left\{\frac{1}{6L},\ \frac{1}{2L\big(3 + 15\,c_0(\tau)/\tau\big)}\right\}, \quad c_0(\tau) := (1-\tau)\left(1 + \frac{2}{\tau}\right).$$

Then the iterates of Top-K EF-SGD satisfy

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\big[\|\nabla f(w_t)\|^2\big] \ \le\ \frac{2\Delta_f}{\eta T} \ +\ 2L\eta\left(3 + \frac{15\,c_0(\tau)}{\tau}\right)\sigma^2 \ +\ \frac{1}{2LT}\,\mathbb{E}\big[\|\nabla f(w_{T+1})\|^2\big].$$

In particular, choosing $\eta = \Theta(1/\sqrt{T})$ obeying the step-size condition gives

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\big[\|\nabla f(w_t)\|^2\big] = O\!\left(\frac{1}{\sqrt{T}}\left[\Delta_f L + \left(3 + \frac{15\,c_0(\tau)}{\tau}\right)\sigma^2\right]\right).$$

Thus Top-K EF-SGD attains the standard nonconvex $O(1/\sqrt{T})$ rate with explicit dependence on the sparsity via $\tau = K/d$.

Theoretical Properties

- Stability and error damping: Error-feedback state satisfies a contraction-type recursion with factor $1 - \tau/2$, ensuring bounded accumulation even for biased compressors. - Sensitivity to K (via τ): Constants scale as $1/\tau$ (and $c_0(\tau) = (1-\tau)\left(1+\frac{2}{\tau}\right) = \frac{2}{\tau} - 1 - \tau \le \frac{2}{\tau}$), capturing the intuition that more aggressive sparsification (smaller K) slows convergence through larger constants. - Comparison to SGD: With $K = d$ ($\tau = 1$), EF-SGD reduces to standard SGD and the bound matches classical results up to constant factors. - Variance-free case: If $\sigma^2 = 0$, the rate simplifies to $\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{2\Delta_f}{\eta T}$ (up to a single-iterate boundary term) with the same step-size condition.

Preliminaries (Definitions and Lemmas)

We compile properties used in the proof.
Definition (Contractive compressor): A mapping $\mathcal{C} : \mathbb{R}^d \to \mathbb{R}^d$ is τ-contractive if

$$\|x - \mathcal{C}(x)\|^2 \le (1-\tau)\|x\|^2 \quad \forall x \in \mathbb{R}^d.$$

Top-K satisfies this with $\tau = K/d$.
Lemma 1 (Top-K contractivity). For $\mathcal{C}(x) = \mathrm{TopK}(x)$, for any $x \in \mathbb{R}^d$,

$$\|x - \mathcal{C}(x)\|^2 \le (1-\tau)\|x\|^2, \quad \tau = K/d.$$

Proof sketch: the K largest-magnitude coordinates of x contain at least a fraction K/d of the squared ℓ_2 norm (worst case is equal magnitudes), hence the residual energy is at most $(1-\tau)$ times the total.
Lemma 2 (Error recursion). Let $u_t = e_t + \eta g_t$, $e_{t+1} = u_t - \mathcal{C}(u_t)$ and assume $\mathcal{C}$ is τ-contractive. Then, for any $\alpha > 0$,

$$\|e_{t+1}\|^2 \le (1-\tau)\|u_t\|^2 \le (1-\tau)\Big((1+\alpha)\|e_t\|^2 + \Big(1 + \tfrac{1}{\alpha}\Big)\eta^2\|g_t\|^2\Big).$$

In particular, with $\alpha = \tau/2$,

$$\|e_{t+1}\|^2 \le \Big(1 - \frac{\tau}{2}\Big)\|e_t\|^2 + c_0(\tau)\,\eta^2\|g_t\|^2,$$

where $c_0(\tau) := (1-\tau)\Big(1 + \frac{2}{\tau}\Big)$.
Lemma 3 (Summation bound for error state). Let $x_{t+1} \le (1-\rho)x_t + a_t$ with $x_1 = 0$, $\rho \in (0,1]$, $a_t \ge 0$. Then

$$\sum_{t=1}^{T+1} x_t \le \frac{1}{\rho}\sum_{t=1}^{T} a_t.$$

Applying with $x_t = \mathbb{E}\|e_t\|^2$, $\rho = \tau/2$, and $a_t = c_0(\tau)\,\eta^2\,\mathbb{E}\|g_t\|^2$, we obtain

$$\sum_{t=1}^{T+1} \mathbb{E}\|e_t\|^2 \le \frac{2c_0(\tau)}{\tau}\,\eta^2\sum_{t=1}^{T}\mathbb{E}\|g_t\|^2.$$

Main Proof (Step-by-Step Derivation)

We prove the main theorem with full detail.

Notation: $d_t := s_t = \mathcal{C}(e_t + \eta g_t)$ and recall $d_t = \eta g_t + e_t - e_{t+1}$.

Start from L-smoothness [Step 1]:

$$f(w_{t+1}) \le f(w_t) + \langle \nabla f(w_t), w_{t+1} - w_t\rangle + \frac{L}{2}\|w_{t+1} - w_t\|^2.$$

Substitute $w_{t+1} - w_t = -d_t$ [Step 2]:

$$f(w_{t+1}) \le f(w_t) - \langle \nabla f(w_t), d_t\rangle + \frac{L}{2}\|d_t\|^2. \tag{1}$$

Express $d_t = \eta g_t + e_t - e_{t+1}$ [Step 3] to obtain

$$-\langle \nabla f(w_t), d_t\rangle = -\eta\langle \nabla f(w_t), g_t\rangle - \langle \nabla f(w_t), e_t - e_{t+1}\rangle. \tag{2}$$

We now sum (1) over $t = 1$ to T and take total expectation. Using (2), we get

$$\mathbb{E}[f(w_{T+1})] \le \mathbb{E}[f(w_1)] - \sum_{t=1}^{T}\eta\,\mathbb{E}\langle \nabla f(w_t), g_t\rangle - \sum_{t=1}^{T}\mathbb{E}\langle \nabla f(w_t), e_t - e_{t+1}\rangle + \frac{L}{2}\sum_{t=1}^{T}\mathbb{E}\|d_t\|^2. \tag{3}$$

Handle the stochastic gradient term via unbiasedness [Step 4]:

$$\mathbb{E}\big[\langle \nabla f(w_t), g_t\rangle \mid \mathcal{F}_t\big] = \langle \nabla f(w_t), \mathbb{E}[g_t \mid \mathcal{F}_t]\rangle = \|\nabla f(w_t)\|^2.$$

Thus

$$\sum_{t=1}^{T}\eta\,\mathbb{E}\langle \nabla f(w_t), g_t\rangle = \eta\sum_{t=1}^{T}\mathbb{E}\|\nabla f(w_t)\|^2. \tag{4}$$

For the error cross-term, write a telescoping decomposition [Step 5]:

$$\langle \nabla f(w_t), e_t - e_{t+1}\rangle = \langle \nabla f(w_t), e_t\rangle - \langle \nabla f(w_t), e_{t+1}\rangle.$$

Add and subtract $\nabla f(w_{t+1})$ in the second inner product [Step 6]:

$$-\langle \nabla f(w_t), e_{t+1}\rangle = -\langle \nabla f(w_{t+1}), e_{t+1}\rangle - \langle \nabla f(w_t) - \nabla f(w_{t+1}), e_{t+1}\rangle.$$

Summing over t and telescoping [Step 7]:

$$\sum_{t=1}^{T}\langle \nabla f(w_t), e_t - e_{t+1}\rangle = \underbrace{\langle \nabla f(w_1), e_1\rangle}_{=0} - \langle \nabla f(w_{T+1}), e_{T+1}\rangle - \sum_{t=1}^{T}\langle \nabla f(w_t) - \nabla f(w_{t+1}), e_{t+1}\rangle. \quad (5)$$

Use Lipschitz gradients to bound the correction term [Step 8]:

$$\|\nabla f(w_t) - \nabla f(w_{t+1})\| \le L\|w_t - w_{t+1}\| = L\|d_t\|.$$

Apply Cauchy-Schwarz and $2ab \le a^2 + b^2$ [Step 9]:

$$\begin{aligned} \big|\langle \nabla f(w_t) - \nabla f(w_{t+1}), e_{t+1}\rangle\big| &\le \|\nabla f(w_t) - \nabla f(w_{t+1})\|\,\|e_{t+1}\| \le L\|d_t\|\,\|e_{t+1}\| \\ &\le \frac{L}{2}\|d_t\|^2 + \frac{L}{2}\|e_{t+1}\|^2. \end{aligned} \quad (6)$$

Substitute (4) and (5)–(6) into (3) [Step 10]:

$$\mathbb{E}[f(w_{T+1})] \le \mathbb{E}[f(w_1)] - \eta\sum_{t=1}^{T}\mathbb{E}\|\nabla f(w_t)\|^2 + \mathbb{E}\langle \nabla f(w_{T+1}), e_{T+1}\rangle + L\sum_{t=1}^{T}\mathbb{E}\|d_t\|^2 + \frac{L}{2}\sum_{t=1}^{T}\mathbb{E}\|e_{t+1}\|^2. \quad (7)$$

We next bound $\|d_t\|^2$ using $(a+b+c)^2 \le 3(\|a\|^2 + \|b\|^2 + \|c\|^2)$ [Step 11]:

$$\|d_t\|^2 = \|\eta g_t + e_t - e_{t+1}\|^2 \le 3\eta^2\|g_t\|^2 + 3\|e_t\|^2 + 3\|e_{t+1}\|^2. \quad (8)$$

Combining (7) and (8) [Step 12]:

$$\begin{aligned} \mathbb{E}[f(w_{T+1})] \le\; &\mathbb{E}[f(w_1)] - \eta\sum_{t=1}^{T}\mathbb{E}\|\nabla f(w_t)\|^2 + \mathbb{E}\langle \nabla f(w_{T+1}), e_{T+1}\rangle \\ &+ L\sum_{t=1}^{T}\mathbb{E}\big[3\eta^2\|g_t\|^2 + 3\|e_t\|^2 + 3\|e_{t+1}\|^2\big] + \frac{L}{2}\sum_{t=1}^{T}\mathbb{E}\|e_{t+1}\|^2. \end{aligned} \quad (9)$$

Group like terms [Step 13]:

$$\begin{aligned} \mathbb{E}[f(w_{T+1})] \le\; &\mathbb{E}[f(w_1)] - \eta\sum_{t=1}^{T}\mathbb{E}\|\nabla f(w_t)\|^2 + \mathbb{E}\langle \nabla f(w_{T+1}), e_{T+1}\rangle \\ &+ 3L\eta^2\sum_{t=1}^{T}\mathbb{E}\|g_t\|^2 + 3L\sum_{t=1}^{T}\mathbb{E}\|e_t\|^2 + \frac{7L}{2}\sum_{t=1}^{T}\mathbb{E}\|e_{t+1}\|^2. \end{aligned} \quad (10)$$

Invoke the second-moment bound from A4: $\mathbb{E}[\|g_t\|^2] \le \mathbb{E}\|\nabla f(w_t)\|^2 + \sigma^2$ [Step 14], which yields

$$3L\eta^2\sum_{t=1}^{T}\mathbb{E}\|g_t\|^2 \le 3L\eta^2\sum_{t=1}^{T}\mathbb{E}\|\nabla f(w_t)\|^2 + 3L\eta^2 T\sigma^2. \quad (11)$$

Next, control the error sums via Lemma 3 [Step 15]. From Lemma 3 with $\rho = \tau/2$:

$$\begin{aligned}\sum_{t=1}^{T+1} \mathbb{E}\|e_t\|^2 &\le \frac{2c_0(\tau)}{\tau}\,\eta^2 \sum_{t=1}^{T} \mathbb{E}\|g_t\|^2 \\ &\le \frac{2c_0(\tau)}{\tau}\,\eta^2 \sum_{t=1}^{T} \Big(\mathbb{E}\|\nabla f(w_t)\|^2 + \sigma^2\Big). \end{aligned} \tag{12}$$

Use (12) to bound the two error sums in (10) [Step 16]. Note that

$$\sum_{t=1}^{T} \mathbb{E}\|e_t\|^2 \le \sum_{t=1}^{T+1} \mathbb{E}\|e_t\|^2, \quad \sum_{t=1}^{T} \mathbb{E}\|e_{t+1}\|^2 = \sum_{t=2}^{T+1} \mathbb{E}\|e_t\|^2 \le \sum_{t=1}^{T+1} \mathbb{E}\|e_t\|^2.$$

Hence

$$\begin{aligned}3L\sum_{t=1}^{T} \mathbb{E}\|e_t\|^2 + \frac{7L}{2}\sum_{t=1}^{T} \mathbb{E}\|e_{t+1}\|^2 &\le \frac{13L}{2}\sum_{t=1}^{T+1} \mathbb{E}\|e_t\|^2 \\ &\le \frac{13Lc_0(\tau)}{\tau}\,\eta^2 \sum_{t=1}^{T} \Big(\mathbb{E}\|\nabla f(w_t)\|^2 + \sigma^2\Big) \\ &= \frac{13Lc_0(\tau)}{\tau}\,\eta^2 \sum_{t=1}^{T} \mathbb{E}\|\nabla f(w_t)\|^2 + \frac{13Lc_0(\tau)}{\tau}\,\eta^2 T\sigma^2. \end{aligned} \tag{13}$$

Substitute (11) and (13) into (10) [Step 17]:

$$\begin{aligned}\mathbb{E}[f(w_{T+1})] \le \mathbb{E}[f(w_1)] &- \eta \sum_{t=1}^{T} \mathbb{E}\|\nabla f(w_t)\|^2 + \mathbb{E}\langle \nabla f(w_{T+1}), e_{T+1}\rangle \\ &+ 3L\eta^2 \sum_{t=1}^{T} \mathbb{E}\|\nabla f(w_t)\|^2 + 3L\eta^2 T\sigma^2 \\ &+ \frac{13Lc_0(\tau)}{\tau}\,\eta^2 \sum_{t=1}^{T} \mathbb{E}\|\nabla f(w_t)\|^2 + \frac{13Lc_0(\tau)}{\tau}\,\eta^2 T\sigma^2. \end{aligned} \tag{14}$$

Rearrange gradient-norm terms to the left [Step 18]:

$$\begin{aligned}\Big(\eta - 3L\eta^2 - \frac{13Lc_0(\tau)}{\tau}\eta^2\Big) \sum_{t=1}^{T} \mathbb{E}\|\nabla f(w_t)\|^2 &\le \mathbb{E}[f(w_1)] - \mathbb{E}[f(w_{T+1})] + \mathbb{E}\langle \nabla f(w_{T+1}), e_{T+1}\rangle \\ &\quad + \Big(3L + \frac{13Lc_0(\tau)}{\tau}\Big)\eta^2 T\sigma^2. \end{aligned} \tag{15}$$

Since $f(w_{T+1}) \ge f^*$ [Step 19], we have

$$\mathbb{E}[f(w_1)] - \mathbb{E}[f(w_{T+1})] \le \Delta_f.$$

Applying Young's inequality with parameter $1/(4L)$ gives

$$\mathbb{E}\langle \nabla f(w_{T+1}), e_{T+1}\rangle \le \frac{1}{4L}\,\mathbb{E}\|\nabla f(w_{T+1})\|^2 + L\,\mathbb{E}\|e_{T+1}\|^2. \tag{16}$$

From Lemma 3 and the fact that $\mathbb{E}\|e_{T+1}\|^2 \le \sum_{t=1}^{T+1} \mathbb{E}\|e_t\|^2$, we have

$$\mathbb{E}\|e_{T+1}\|^2 \le \sum_{t=1}^{T+1} \mathbb{E}\|e_t\|^2 \le \frac{2c_0(\tau)}{\tau}\,\eta^2 \sum_{t=1}^{T} \mathbb{E}\|g_t\|^2 \le \frac{2c_0(\tau)}{\tau}\,\eta^2 \sum_{t=1}^{T} \Big(\mathbb{E}\|\nabla f(w_t)\|^2 + \sigma^2\Big). \tag{17}$$

Combine (15), (16), and (17), and group gradient terms to the left:

$$\begin{aligned}\Big(\eta - 3L\eta^2 - \frac{13Lc_0(\tau)}{\tau}\eta^2\Big) \sum_{t=1}^{T} \mathbb{E}\|\nabla f(w_t)\|^2 \le \Delta_f + \frac{1}{4L}\,\mathbb{E}\|\nabla f(w_{T+1})\|^2 \\ + L \cdot \frac{2c_0(\tau)}{\tau}\,\eta^2 \sum_{t=1}^{T} \mathbb{E}\|\nabla f(w_t)\|^2 \\ + \left(3L + \frac{13Lc_0(\tau)}{\tau}\right)\eta^2 T\sigma^2 + L \cdot \frac{2c_0(\tau)}{\tau}\,\eta^2 T\sigma^2.\end{aligned} \tag{18}$$

Move the gradient-sum term from the right-hand side to the left to obtain

$$\Big(\eta - L\eta^2\big(3 + \tfrac{15c_0(\tau)}{\tau}\big)\Big) \sum_{t=1}^{T} \mathbb{E}\|\nabla f(w_t)\|^2 \le \Delta_f + \frac{1}{4L}\,\mathbb{E}\|\nabla f(w_{T+1})\|^2 + L\eta^2\Big(3 + \tfrac{15c_0(\tau)}{\tau}\Big)T\sigma^2. \tag{19}$$

Step-size condition selection. Assume

$$\eta \le \min\left\{\frac{1}{6L},\ \frac{1}{2L\Big(3 + \frac{15c_0(\tau)}{\tau}\Big)}\right\}$$

so that $L\eta\Big(3 + \frac{15c_0(\tau)}{\tau}\Big) \le \frac{1}{2}$. Then

$$\eta - L\eta^2\Big(3 + \tfrac{15c_0(\tau)}{\tau}\Big) \ge \frac{\eta}{2}.$$

Applying this to (19) yields [Step 26]

$$\frac{\eta}{2} \sum_{t=1}^{T} \mathbb{E}\|\nabla f(w_t)\|^2 \le \Delta_f + \frac{1}{4L}\,\mathbb{E}\|\nabla f(w_{T+1})\|^2 + L\eta^2\Big(3 + \tfrac{15c_0(\tau)}{\tau}\Big)T\sigma^2. \tag{20}$$

Divide by ηT to average [Step 27]:

$$\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{2\Delta_f}{\eta T} + \frac{1}{2LT}\,\mathbb{E}\|\nabla f(w_{T+1})\|^2 + 2L\eta\Big(3 + \tfrac{15c_0(\tau)}{\tau}\Big)\sigma^2. \tag{21}$$

This completes the proof.

Rate Derivation (With Constants)

Set $\eta = \min\Big\{\eta_0,\ \frac{c}{\sqrt{T}}\Big\}$ with

$$\eta_0 := \min\left\{\frac{1}{6L},\ \frac{1}{2L\big(3 + 15c_0(\tau)/\tau\big)}\right\}, \quad c > 0.$$

Then from (21):

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{2\Delta_f}{\eta T} + \frac{1}{2LT}\mathbb{E}\|\nabla f(w_{T+1})\|^2 + 2L\eta\left(3 + \frac{15c_0(\tau)}{\tau}\right)\sigma^2.$$

If $\eta = c/\sqrt{T}$ (and T large so that $\eta \le \eta_0$), we obtain

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{2\Delta_f}{c}\cdot\frac{1}{\sqrt{T}} + O\left(\frac{1}{T}\right) + \left[2Lc\left(3 + \frac{15c_0(\tau)}{\tau}\right)\right]\frac{\sigma^2}{\sqrt{T}} = O\left(\frac{1}{\sqrt{T}}\right),$$

with explicit constants depending on $L, \sigma^2, \Delta_f, \tau$.

Special Cases

- No compression ($K = d$, $\tau = 1$): $c_0(1) = 0$. The bound reduces to

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{2\Delta_f}{\eta T} + 6L\eta\,\sigma^2 + \frac{1}{2LT}\mathbb{E}\|\nabla f(w_{T+1})\|^2,$$

matching the classical SGD rate (up to constants). - Variance-free gradients ($\sigma^2 = 0$): The bound simplifies to

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{2\Delta_f}{\eta T} + \frac{1}{2LT}\mathbb{E}\|\nabla f(w_{T+1})\|^2,$$

so with $\eta = \eta_0$ fixed we get $O(1/T)$ decay in the average squared gradient norm up to a single-iterate boundary term. - Extreme sparsity (small τ): Constants scale like $1/\tau$ through $c_0(\tau)$ and $A(\tau) := 3+15c_0(\tau)/\tau$, reflecting the increased error due to aggressive sparsification. Error-feedback ensures stability via the $(1-\tau/2)$ contraction in Lemma 2.